wire fraud, paid $1.8 billion in fines, and stopped managing money for outside investors.[47])

Finally, HFs can impose *gates*, which allow the HFs to limit redemptions, and sometimes redemptions can be *suspended* altogether.

Lockups allow HF managers to pursue strategies that would be difficult if investors withdrew at the wrong time. That is, selling volatility will earn a long-run risk premium but only if you stick through the times when volatility spikes and prices crash. Lockups allow funds to hold more illiquid assets and earn an illiquidity risk premium (see chapter 13). Aragon (2007) shows that HFs with lockup restrictions earn 4% to 7% higher returns than HFs without lockups.

Gates and suspensions cause HFs to be like the "Hotel California" in the song by the Eagles. Investors may request their funds ("you can check out anytime you like") but HFs won't give your money back ("but you can never leave"). During the financial crisis, many investors wanted to check out of badly performing HFs, but they were stuck. This was not just a problem for 2008–2009; HFs get in trouble all the time and gate. In October 2012, the Endowment Fund, a multibillion HF run by Mark Yusko, the former chief of the endowment of the University of North Carolina at Chapel Hill, imposed gates after lousy performance over the prior two years.[48]

Illiquidity is costly to the asset owner. In Ang and Bollen (2010a, 2010b), I compute the cost of lockup and redemption restrictions using a real option approach.[49] An investor gives up an option—the right to get out of the fund when she wants, and she most wants to exercise the option when the manager is destroying value. This is precisely the time that the HF doesn't allow you to leave. I estimate the cost of a three- to six-month lockup at around 2%. The cost of a two-year lockup is 4%. The cost of illiquidity increases to 15% if the HF suspends all redemptions during bad times and rescinds the liquidity option when the investor desires it most.

4.4. THE FUTURE

HFs are not going away, and some generate sizeable returns. But most HFs are just conduits of risk factors, in opaque, illiquid, and expensive wrappers. Many of these risk factors—like volatility, credit, small-large, value-growth, and momentum—are especially attractive for long-term investors. Is there a better way of accessing these risk premiums?

The barrier to entry for many factor strategies is low. There are now mutual funds that do HF-like strategies. Some of them are HFs in everything but name—but with the protections that the 40-Act confers to investors. A few of these

[47] Protess, B., and P. Lattman, "After a Decade, SAC Capital Blinks," *New York Times*, Nov, 4, 2013.

[48] See Creswell, J., "After Weak Returns, the Endowment Fund Limits Withdrawals," *New York Times*, Oct. 30, 2012.

[49] See also Derman (2007).